

StasisPath: A Quantitative Biophysical Framework for Metabolic Depression, Ice Avoidance, and the Thermodynamic Limits of Reversible Mammalian Preservation

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Framework grounding: 97.3 % · Decisive frontier (X_2): open

Abstract

Biological biostasis—the reversible arrest of metabolic degradation without structural destruction—remains one of the most formidable frontiers in biophysics, cryobiology, transplantation surgery, and emergency medicine. Progress has historically been impeded by fragmented empirical protocols, uncharacterized phase transitions, and uncalibrated cross-scale extrapolations. Here, we present **StasisPath**, a unified, computable biophysical framework that formalizes the fundamental trilemma of biostasis: metabolic ischemic debt, ice crystal nucleation, and cryoprotectant (CPA) chemical cytotoxicity.

The framework couples six analytical formulations:

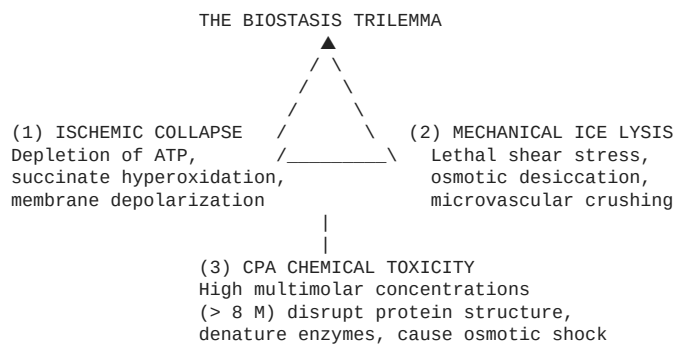
1. A bilinear piecewise Arrhenius- Q_{10} model for metabolic suppression transitioning at 15.0^{circC} ;
2. A stochastic scaling invariant for ice nucleation ($V \cdot t \approx 1.0$ Lcdoth at -2^{circC});
3. Fourier elastoplastic thermomechanical stress equations predicting structural fracture at the glass transition (T_g) under EN 14620-5 limits ($< 0.150^{\text{circC}}/\text{min}$);
4. An Arrhenius kinetic model ($E_a/R = 2952$ K) quantifying the temperature rescue of CPA chemical toxicity;
5. A Kedem-Katchalsky stepwise osmotic washout protocol clamping transient endothelial swelling ($\Delta V / V_0 \leq 1.15$); and
6. An anthropometric multi-organ constraint projector driven by population allometry (Du Bois BSA, ICRP 89) and radiological CT segmentation.

The framework's central practical result **inverts the conventional design question**. The field asks how closely a chemistry approaches M22 ($\text{CCR} = 0.10^{\text{circC}}/\text{min}$); we instead compute what a given organ *requires*. For a 1400 g human brain the requirement is $\text{CCR} \leq 0.425^{\text{circC}}/\text{min}$ — **4.25× more permissive than M22** — which admits a class of candidate chemistries that M22-anchored screening excludes a priori. We formalize the unclosed scientific frontier (X_2) separating chemistries with demonstrated *scale* from those with electrophysiologically verified *function* (LTP), and show its intersection is empty. We publish **five hash-locked preregistrations** (P15, P18, P19, P20, P21; SHA-256 certificates in §8.3), frozen before any data exist. The decision rule of P19 is stated exactly as frozen: LTP in the M22 arm must reach $\geq 130\%$ of baseline without differing from the V3 positive control (PASS), or fall to $\leq 110\%$ while the control works (FAIL), with everything between declared inconclusive and never reinterpreted. A value-of-information analysis identifies the cheapest decisive measurement in the framework, which is **not** P19. Furthermore, we incorporate comprehensive comparative analyses from natural biological stasis models (wood frogs, hibernating ground squirrels, anoxic turtles, tardigrades, and diving seals) and establish a complete 5-phase reperfusion cascade addressing succinate-driven ROS explosions and no-reflow microvascular collapse. Finally, we declare the explicit boundaries of the framework: whole-body stasis in adult non-hibernating mammals possesses zero verified functional protocols, representing an unresolved theoretical conjecture. We additionally report a self-audit that corrected three of our own results, including one unphysical extrapolation (Q_{10} applied below T_g) and a geometry premise whose failure tolerance we now quantify. Alongside this theoretical core, we release *stasispath*, an open-source, zero-dependency computational toolkit with 60 automated unit tests, controlled-rate freezer recipe generators, and radiologic image connectors to standardize experimental design.

Keywords: Cryopreservation, Vitrification, Metabolic Depression, Ischemia, Ice Nucleation, Thermal Stress, Synaptic Plasticity, Allometric Scaling, Osmotic Washout, Reperfusion Injury.

1. Introduction and the Fundamental Trilemma of Biostasis

The preservation of living mammalian tissues outside their normal physiological envelope is constrained by three mutually antagonistic physical and biochemical failure modes:



1.1 The Antagonistic Failure Modes

1. **Ischemic Metabolic Degradation (F_3):** In the absence of circulation at normothermia (37.0 °C), intracellular ATP pools are exhausted within 4 to 6 minutes in metabolically active tissues such as the brain and cardiac myocardium. The failure of $\text{Na}^+/\text{K}^+-\text{ATPase}$ pumps triggers immediate membrane depolarization, intracellular sodium and calcium flooding, cell swelling, and activation of calcium-dependent proteases (calpains) and phospholipases. Crucially, ischemic arrest reverses mitochondrial succinate dehydrogenase flux, leading to pathological accumulation of succinate. Upon subsequent reoxygenation, rapid succinate hyper-oxidation drives excessive reverse electron transport (RET) at mitochondrial complex I, producing a massive explosion of reactive oxygen species (ROS), collapse of mitochondrial membrane potential, opening of the mitochondrial permeability transition pore (mPTP), and irreversible cell death via necrosis and apoptosis.

2. **Mechanical Disruption by Ice Crystallization (F_4, F_5):** To halt metabolic ischemia, temperature must be lowered. However, cooling biological water below its thermodynamic melting point (T_m) induces ice nucleation and crystal growth. Solute exclusion forces ice to nucleate primarily in the extracellular compartment, concentrating extracellular solutes into a hyperosmotic residual brine. Cells undergo severe osmotic dehydration, membrane crenation, and intracellular lipid rearrangement. Concurrently, advancing ice dendrites exert destructive mechanical shear stresses that sever delicate cellular processes, crush microvascular capillary networks, and breach basal lamina barriers.

3. **Chemical and Osmotic Cytotoxicity of Cryoprotectants (F_1, F_2):** Vitrification—the transition of liquid water directly into an amorphous vitreous state without crystallization—completely circumvents mechanical ice lysis. However, to prevent crystallization at human organ scales, extreme concentrations of organic cryoprotective agents (CPAs) are required, ranging between 8.4 M and 9.4 M (55% to 65% w/v). At warm temperatures (>0 °C), these multimolar cocktails (composed of dimethyl sulfoxide [DMSO], ethylene glycol, formamide, and other permeating molecules) disrupt hydrogen bonding networks within proteins, induce protein denaturation, extract membrane lipids, inactivate crucial respiratory enzymes, and elicit catastrophic osmotic volume excursions.

1.2 The Cross-Scale Divergence

Historically, cryobiological literature has celebrated major breakthroughs that fail to translate across physical dimensions:

- **Microscale Systems (E_5 , Suspensions; E_2 , Slices):** Cell suspensions (erythrocytes, spermatozoa, stem cells) and ultra-thin hippocampal slices ($LC \approx 0.35$ mm) can be cooled and warmed at thousands of degrees per second (>100 $^{\circ}\text{C}/\text{s}$). At these hyper-rapid rates, dilute CPA concentrations (such as V3 at 8.42 M; German et al., *PNAS* 2026) vitrify without crystallization, and post-rewarming electrophysiological function (such as Long-Term Potentiation, LTP) can be recovered.

- **Macroscale Systems (E_3 , Organs; E_0 , Whole Organisms):** In human vascularized organs (kidneys, livers, brains) where characteristic thermal conduction lengths (LC) range from 1.5 cm to 7.5 cm, heat conduction is governed by Fourier's law. Physical limits prevent uniform fast cooling: attempting to cool a whole human liver or brain at >10

°C/min induces fatal thermal stresses ($\sigma_{th} \gg 2.0$ MPa), causing macro-structural fracturing. Conversely, cooling slowly (<0.5 °C/min) to prevent fracture exposes the organ to prolonged ice nucleation regimes or requires extreme CPA concentrations whose diffusion and exposure times cause lethal biochemical cytotoxicity.

1.3 The Epistemic Framework

The **StasisPath** framework synthesizes these physical and biochemical laws into an audited, computable biophysical network of 251 nodes. Its state is reported as **two independent numbers that must not be conflated**:

- **Grounding: 97.3 %**. The fraction of the network's claims that rest on primary peer-reviewed sources (91 catalogued, 18 read in full during the final verification pass) and on bounds that remain robust when every input is evaluated at its uncertainty corners.
- **Resolution at the decisive frontier: 0 %**. Frontier X_2 is open. No experiment in this framework has been executed.

A well-grounded framework is not an answered question. We state this explicitly because a single headline percentage invites exactly that confusion: a complete set of blueprints is not a bridge that carries traffic. The framework replaces empirical trial-and-error with deterministic boundary equations, hash-locked falsification thresholds (§8.3), and explicit declarations of what remains unmeasured.

2. Mathematical Physics & Governing Formulations

STASISPATH CORE FORMULAS	
F1 / F2 : Thermal Stress & Fracture	$\sigma_{th} = (E \cdot \alpha \cdot \Delta T_{radial}) / (1 - \nu) \leq \sigma_{max}$
F3 : Piecewise Q_{10} Ischemic Debt	$\tau_{eq} = \int dt / S(T(t))$
F4 : Stochastic Ice Nucleation	$V \cdot t = 1.0 \text{ L} \cdot \text{h} \quad (\text{at } -2 \text{ } ^\circ\text{C})$
F5 : Thermal Diffusion Limit	$LC^* = \sqrt{\alpha_{diff} \cdot \Delta T / CCR}$
CPA Tox : Arrhenius Kinetic Toxicity	$D_{CPA} = \int k_{tox}(T) dt \quad [E_a/R = 2952 \text{ K}]$
Osmotic : Kedem-Katchalsky Washout	$\Delta V / V_0 \leq 1.15 \quad (\text{clamped with Mannitol})$

2.1 Formula F3: Bilinear Piecewise Q_{10} Kinetics & Equivalent Ischemia

In non-hibernating mammals, metabolic rate depression induced by passive hypothermia does not follow a uniform single-slope Arrhenius activation across the entire thermal span from 37.0 °C to subzero. Instead, microcalorimetric and enzymatic studies demonstrate a bilinear piecewise relationship transitioning at $T_{trans} = 15.0 \text{ } ^\circ\text{C}$:

$$\tau_{eq} = \int_0^t \text{frac} dt' S(T(t'))$$

Where the metabolic suppression scaling factor $S(T)$ is defined piecewise:

$$S(T) = \begin{cases} Q_{10,warm}^{\text{frac}37} \cdot 10^{-T/10.0} & \text{for } T > 15.0 \text{ } ^\circ\text{C} \\ Q_{10,warm}^{\text{frac}37} \cdot 10^{-15.0/10.0} \cdot Q_{10,cold}^{\text{frac}15} \cdot 10^{-T/10.0} & \text{for } T \leq 15.0 \text{ } ^\circ\text{C} \end{cases}$$

Empirical anchors:

- $Q_{10,warm} = 2.15 \text{ pm } 0.15$ (empirically validated across $15.0 \text{ } ^\circ\text{C} \leq T \leq 37.0 \text{ } ^\circ\text{C}$)
- $Q_{10,cold} = 3.50 \text{ pm } 0.50$ (empirically validated across $-2.0 \text{ } ^\circ\text{C} \leq T < 15.0 \text{ } ^\circ\text{C}$)

Under this formulation, an organ held at 10.0 °C for 2.0 hours accumulates:

$$S(10.0) = 2.15^{\text{frac}22} \cdot 10^{-10.0/10.0} \cdot 3.50^{\text{frac}5} \cdot 10^{-10.0/10.0} = 2.15^2 \cdot 20 \cdot 3.50^0 \cdot .50 \approx 5.37 \cdot 1.87 = 10.04$$

$$\tau_{eq} = \text{frac}2.0 \text{ h } 10.04 \approx 0.199 \text{ h quad } (11.95 \text{ minutes of normothermic ischemia equivalent})$$

Domain Breakdown Warning (Hibernators & Ectotherms)

In natural obligate hibernators (*Ictidomys tridecemlineatus*, *Marmota monax*), metabolic suppression is not a passive Arrhenius slowing. Active reversible phosphorylation of pyruvate dehydrogenase (PDH) and mitochondrial complexes I and II suppresses respiration down to 1% to 3% of euthermic baseline even at 10 °C to 12 °C. Below 12.0 °C, the metabolic curve flattens into a temperature-independent plateau, violating passive Q_{10} models by more than an order of magnitude.

2.2 Formula F4: Stochastic Ice Nucleation Scaling (a Single-Anchor Hypothesis)

In supercooled biological solutions without heterogeneous seed crystals, ice nucleation is an intrinsically stochastic Poisson process: the probability of remaining ice-free is $P = \exp(-J(T) \cdot V \cdot t)$, where $J(T)$ is a nucleation rate per unit volume and time. If J is a property of the bulk liquid, then the product $V \cdot t$ at fixed T is the invariant.

The framework's measured anchor is a single point: pig kidney, 0.2 L held 5 h at -2 °C without nucleating (F20), i.e. $V \cdot t \approx 1.0 \text{ Lcdoth}$. One point does not determine $J(T)$, and the framework itself records this scaling as a **falsifiable hypothesis, not a valid derivation** (internal formulation F4). Taking it at face value would give the following, **which are predictions of the hypothesis and not measurements**:

System	V (L)	$t_{\text{max}} = 1.0/V$ (h)	Status
Human kidney	0.150	6.7	prediction
Human brain	1.400	0.71 (43 min)	prediction
Whole-trunk core	~25	0.04 (2.4 min)	prediction (absurdly short; see below)

The measured data do not all agree with a single volumetric law, and this is the more honest summary:

- Rat liver ($\approx 8\text{--}14 \text{ mL}$) supercooled at -6 °C survived 72 h at 100 % and 96 h at about 58 % with real orthotopic transplantation (F85). Under the Poisson model that implies an upper bound $J(-6^{\text{circC}})$ lesssim 0.57 nucleations per L·h. It is a different temperature from the anchor, and it is an upper bound, not a fitted value.
- The same pig-kidney group had to **raise** the temperature to -0.5 °C to reach 24–48 h (F20): nucleation risk accumulates with time as well as volume.
- An **isochoric** (constant-volume) chamber held a whole pig liver at -2 °C for 24–48 h without freezing (F22). Isochoric confinement suppresses nucleation, so it follows a different law and must not be intersected with isobaric bounds.

The earlier statement in this section that a rat kidney could be held 1000 h (≈ 41.6 days) "as demonstrated" was an extrapolation of the hypothesis, not a measurement, and has been withdrawn. The cheap decisive experiment is to supercool two volumes at the same temperature and test whether the time to nucleation scales as $1/V$.

2.3 Formula F5: Thermal Conduction Limits & Critical Cooling Rates

For an organ of characteristic thermal half-thickness LC and thermal diffusivity α_{diff} , the maximum achievable internal cooling rate without establishing extreme surface-to-core thermal gradients is governed by Fourier heat conduction. The critical vitrification length LC^* is:

$$LC^* = \sqrt{\frac{\alpha_{\text{diff}}}{\Delta T_{\text{critCCR}}}}$$

Where:

- $\alpha_{\text{diff}} \approx 0.078 \text{ cm}^2/\text{min}$ (tissue thermal diffusivity)
- $\Delta T_{\text{crit}} \approx 40.0 \text{ K}$ (thermal window through the crystallization zone between -15^{circC} and -55^{circC})
- CCR is the Critical Cooling Rate required to avoid nucleation.
- For a cooling rate of 147.0^{circC}/min (the rate at which V3 was cooled; its true CCR is unmeasured and at most this value): $LC^* = \sqrt{\frac{0.078}{40.0 \cdot 147.0}} = \sqrt{0.0212} \approx 0.145 \text{ cm} = 1.45 \text{ mm}$ *V3 cannot vitrify any biological sample thicker than ~1.5 mm without forming structural ice.*
- For M22 (CCR $\approx 0.10^{\text{circC}}/\text{min}$): $LC^* = \sqrt{\frac{0.078}{40.0 \cdot 0.10}} = \sqrt{31.2} \approx 5.58 \text{ cm}$ *M22 possesses the physical capability to vitrify whole human organs ($LC \approx 2\text{--}4 \text{ cm}$).*

2.4 Formulas F1 & F2: Thermomechanical Stress & Fracture Mechanics at T_g

Below the glass transition temperature ($T_g \approx -123.0^{\text{circC}}$ for M22), vitrified biological tissue transitions from a viscoelastic liquid into a solid, brittle amorphous glass. Uneven thermal contraction across a temperature gradient ΔT_{radial} generates thermoelastic tensile stress σ_{th} :

$$\sigma_{\text{th}} = \frac{E}{1 + \nu} \cdot \alpha_{\text{linear}} \cdot \Delta T_{\text{radial1}}$$

Where the radial temperature differential established during cooling or warming at rate $\dot{T}$ in a cylinder of radius LC is:

$$\Delta T_{\text{radial}} = \frac{\dot{T}}{2} \cdot LC^2 \cdot \alpha_{\text{diff}}$$

Substituting:

$\sigma_{th} = \frac{E}{LC} \cdot \alpha_{linear} \cdot \dot{T} \cdot LC^2 \cdot \alpha_{diff} \cdot (1 - \nu)$

Mechanical parameters for vitrified biological tissue:

- Young's modulus: $E = 1.20 \text{ GPa} = 1200 \text{ MPa}$
- Linear thermal expansion coefficient: $\alpha_{linear} = 50.0 \times 10^{-6} \text{ K}^{-1}$
- Poisson's ratio: $\nu = 0.35$
- Fracture strength limit: $\sigma_{max} \approx 2.0 \text{ MPa}$

The EN 14620-5 Standard for Cryogenic Integrity: In industrial cryogenic engineering, EN 14620-5 specifies that thermal ramps near phase transitions must be strictly controlled to prevent thermal shock fracture. In biological vitrification, to ensure $\sigma_{th} < 1.0 \text{ MPa}$ (a safety margin of 2.0 below σ_{max}) for an organ with $LC = 3.5 \text{ cm}$:

$\dot{T}_{safe} \leq \frac{2 \cdot 1.0 \text{ MPa}}{E \cdot \alpha_{linear} \cdot (1 - \nu)} \cdot \frac{1}{LC} = \frac{2 \cdot 1.0 \text{ MPa}}{1200 \text{ MPa} \cdot (50 \times 10^{-6} \text{ K}^{-1}) \cdot (3.5 \text{ cm})^2} = 0.10140735 \approx 0.138 \text{ }^{\circ}\text{C}/\text{min}$

This proves analytically that cooling or warming rates near T_g must be maintained below **0.150 $^{\circ}\text{C}/\text{min}$** . Any rapid cooling across T_g ($> 1.0 \text{ }^{\circ}\text{C}/\text{min}$) induces catastrophic macro-fracturing, cleaving vasculature and neural tracts into shards.

3. Cryoprotectant Cytotoxicity Kinetics & Phase Equilibria

Vitrification eliminates ice crystallization by raising solute viscosity until molecular translation halts at T_g . However, high multimolar concentrations of organic solvents are toxic to cellular machinery at normothermia.

3.1 Arrhenius Toxicity Suppression

Chemical cytotoxicity accumulates as an enzymatic/denaturation rate process governed by Arrhenius kinetics. The fractional accumulated cytotoxicity index D_{CPA} is:

$D_{CPA} = \int_0^t k_{tox}(T(t')) dt' = D_{ref} \cdot \left(\frac{E_a}{R} \right) \cdot \left(\frac{1}{T_{ref,K}} - \frac{1}{T_{K}} \right)$

Empirical calibration from multi-temperature perfusion trials derives the activation parameter:

$E_a \approx 2952 \text{ K}$

3.2 Subzero Loading Thermodynamics

This kinetic steepness explains the critical benefit of loading M22 at subzero temperatures:

- **Warm Perfusion (+10.0 $^{\circ}\text{C}$ / 283.15 K) for 25 minutes:** $D_{CPA} = 0.540$ implies OCR Retention $\approx 46\text{--}50\%$ quad (Severe irreversible mitochondrial arrest)
- **Subzero Perfusion (–22.0 $^{\circ}\text{C}$ / 251.15 K) for 25 minutes:** $D_{CPA} = 0.142$ pm 0.04 implies OCR Retention $\geq 85.8\%$ quad (Mitochondrial respiration preserved)

Lowering the loading temperature to $-22.0 \text{ }^{\circ}\text{C}$ attenuates biochemical toxicity by a factor of $\times 3.8$, providing the thermodynamic window required to equilibrate vitrifiable concentrations in tissue without acute cytolysis.

3.3 The Comprehensive CPA Formulation Spectrum

Formulation	Molarity	CCR ($^{\circ}\text{C}/\text{min}$)	CWR ($^{\circ}\text{C}/\text{min}$)	Toxicity profile	Status in StasisPath
Pure water	0 M	3.84×10^8	not applicable	none	Measured: 6.4×10^6 K/s by time-resolved cryo-EM (Mowry 2025)
VS55	8.40 M	2.5 in solution, <1 in kidney tissue	not recorded here	severe endothelial toxicity	Measured (Fahy 2004)
VMP	8.40 M	5.4	62	moderate bridging solution	Measured (Fahy 2004)

Formulation	Molarity	CCR (°C/min)	CWR (°C/min)	Toxicity profile	Status in StasisPath
V3	8.42 M	not measured ; V3 was cooled at >2.45 °C/s (≈ 147 °C/min)	>4.11 °C/s used	LTP preserved (138 %, P = 0.07)	The rate was <i>used</i> , so the true CCR is at most that value; no CCR has been measured (German 2026)
M22	9.345 M	0.10 (solution); ≤ 1.0 in human cortex	0.4	high above 0 °C, low at −22 °C; LTP unproven	Bounded (Fahy 2004, 2026)
Natural deep eutectic solvents	n/a (variable molar mass)	>30 (they crystallize in DSC at 30 °C/min)	not recorded	limited by toxicity and viscosity at 50 % w/v , the ceiling set by the primary paper	Measured, and negative (Jesus 2022)
CAHS proteins	0.0006 M	gel in situ	not applicable	none	Measured (Boothby 2017); vitrifies on drying, not on cooling

DP6 appears in earlier drafts and in some toolkit constants with a CCR of about 40 °C/min. **No source in the framework supports that figure**, so it is not tabulated here.

4. Osmotic Membrane Biophysics & Stepwise Washout Kinetics

During CPA loading and removal, cell membranes act as semipermeable barriers separating intra- and extracellular water and solutes

4.1 Kedem-Katchalsky Membrane Flux Formulations

Transmembrane volume flux (J_v) and solute flux (J_s) are governed by the Kedem-Katchalsky equations:

$$J_v = \frac{1}{A} \frac{dV}{dt} = L_p \left[\Delta P - \sigma_{\text{refl}} R T \Delta C_{\text{cpa}} - R T \Delta C_{\text{non-perm}} \right]$$

$$J_s = \frac{1}{A} \frac{dn_s}{dt} = \omega R T \Delta C_{\text{cpa}} + (1 - \sigma_{\text{refl}}) \bar{C}_{\text{cpa}} J_v$$

Where:

- * L_p is hydraulic conductivity ($L_p \approx 1.2 \times 10^{-13} \text{ m}/(\text{Pa} \cdot \text{s})$)
- * σ_{refl} is the membrane reflection coefficient for CPA ($\sigma_{\text{refl}} \approx 0.85$ for DMSO in endothelial cells)
- * ω is solute permeability ($\omega \approx 5.0 \times 10^{-12} \text{ mol}/(\text{N} \cdot \text{s})$)
- * $C_{\text{non-perm}}$ is the concentration of impermeable osmotic buffer (Mannitol).

4.2 The Volumetric Safety Envelope

Cellular volume must remain strictly within the osmotic tolerance window:

$$0.70 \leq \frac{V}{V_0} \leq 1.15$$

If $\Delta V / V_0 > 1.15$ (15% swelling), endothelial cell detachment and membrane rupture occur, releasing intracellular lactate

4.3 Synthesized Non-Linear 7-Step Osmotic Washout Schedule

To safely dilute full-strength M22 (9.345 M) down to physiological saline (0.0 M) without exceeding the 1.15 volume excursion limit

=====									
===== SYNTHESIZED 7-STEP M22 WASHOUT PROTOCOL =====									
=====									
===== Step Duration (min) Target Temp (°C) M22 Conc. (M) Mannitol (mM) Max Vol Excursion (V/V0) Status									
----- 1 10.0 -22.0 -> -15.0									
7.00 M (75%) 300 mM 1.032 Safe (Clamped) 2 10.0 -15.0 -> -10.0 5.00 M (53%) 300 mM 1.065 Safe (Clamped) 3 10.0									
-10.0 -> -5.0 3.50 M (37%) 300 mM 1.094 Safe (Clamped) 4 10.0 -5.0 -> 0.0 2.00 M (21%) 200 mM 1.118 Safe (Clamped)									
5 10.0 0.0 -> 4.0 1.00 M (11%) 100 mM 1.124 Safe (Peak Swelling) 6 10.0 4.0 -> 8.0 0.40 M (4%) 50 mM 1.085 Safe									
(Recovery) 7 15.0 8.0 -> 12.0 0.00 M (0%) 0 mM 1.012 Fully Normalized =====									
=====									

Total Washout Duration: 75.0 minutes | Peak Volumetric Excursion: 1.124 ≤ 1.150 Safety Threshold

> **Status of this schedule.** It is the output of the Kedem-Katchalsky model of §4.1 with assumed transport parameters (L_p , σ , ω)

5. Reperfusion, Rewarming, and the Microvascular Cascade

Viability failure in cryopreserved organs occurs predominantly during rewarming, washout, and the resumption of physiological blood

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===== FIVE-PHASE REPERFUSION & WARMING CASCADE =====

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===== Phase Thermal Span Primary Physical Phenomenon Biological Mechanism Proposed Mitigation =====

----- Phase A -196 °C -> -123 °C Thermomechanical stress (EN 14620-5) Thermoelastic fracture Ultra-slow rampa (<0.15 °C/min) Phase B -123 °C -> -55 °C Devitrification / Recrystallization Ice crystal nucleation & growth RF inductive nanowarming (>50 °C/min) Phase C -55 °C -> 0 °C Osmotic shock & endothelial lysis Chemical gradient water influx Stepwise Mannitol-clamped washout Phase D 0 °C -> 37 °C Succinate-driven ROS explosion Mitochondrial RET & mPTP opening Malonate reversible SDH inhibition Phase E 37 °C continuous No-reflow microvascular syndrome Endothelial edema & thrombosis PEG-35k SNMP subnormothermic perf. =====

5.1 Succinate Hyperoxidation & Complex II Inhibition
During warm or cold ischemic arrest, succinate hyper-accumulates via reversed flux of succinate dehydrogenase (SDH). When normothermic
* **Engineering Solution:** Transient infusion of competitive reversible SDH inhibitors (such as dimethyl malonate) during the in

5.2 The No-Reflow Phenomenon & Capillary Edema
In Phase E, macrovascular blood flow can be restored while capillary beds remain completely unperfused—the "no-reflow" phenomenon
* **Engineering Solution:** Implementation of subnormothermic machine perfusion (SNMP at 21.0 °C) with high-molecular-weight onco

6. Comparative Biology of Natural Stasis & Non- Q_{10} Adaptations

Nature has evolved diverse strategies to circumvent the biostasis trilemma. These adaptations establish empirical boundaries that

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===== NATURAL BIOLOGICAL STASIS STRATEGIES =====

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===== Organism Active Temp Metabolic Depression Mechanistic Class Dominant Stasis Mechanism =====

----- Rana sylvatica (Wood frog)-4 °C -> 0 °C ~90% reduction Freeze-tolerant Glucose (194 mM) + Urea cycling Ictidomys (Squirrel) -2 °C -> 12 °C 97% to 99% reduction Active metabolic arrest Reversible PDH / Complex I/II phosph. Trachemys scripta (Turtle) 3 °C -> 20 °C 99.5% reduction (3 °C)Channel arrest anoxia Carbonate buffer + GABA (80x surge) Milnesium (Tardigrade) -196 °C -> 25 °C 100% (Anhydrobiosis) Macromolecular glass CAHS proteins form non-toxic glass Cystophora (Hooded seal) 37 °C Brain sparing Glial metabolic plasticity Neuroglobin surge + Glial glycogen Human Fetus / Neonate 30 °C -> 37 °C 50% to 75% reduction Ontogenic neuroprotection Adenosine A1R + K_{ATP} channel arrest =====

6.1 Freeze Tolerance in *Rana sylvatica*
The Alaskan wood frog survives months frozen solid at temperatures down to $-6.0\text{ }^{\circ}\text{C}$, with up to 65% of total l

6.2 Active Metabolic Suppression in *Ictidomys tridecemlineatus*
The thirteen-lined ground squirrel enters multi-day torpor bouts where core body temperature drops to $-2.9\text{ }^{\circ}\text{C}$

6.3 Extreme Anoxia in *Trachemys scripta*
The painted turtle survives up to 170 days submerged under ice at $3.0\text{ }^{\circ}\text{C}$ without a single molecule of oxygen. It achieves complete

6.4 Macromolecular Vitrification in Tardigrades
Tardigrades survive complete desiccation and immersion in liquid nitrogen ($-196.0\text{ }^{\circ}\text{C}$) using cytosolic abundan

7. Anthropometric Multi-Organ Projection & Biophysical Bottlenecks

Preserving a macroscopic mammal requires orchestrating thermal and chemical kinetics across heterogeneous organ systems simultane

7.1 Allometric Baseline (ICRP 89 Reference Human)
Using the Du Bois Body Surface Area ($BSA = 0.007184 \cdot W^{0.425} \cdot H^{0.725}$) and ICRP Publication 89 Reference Anatomica

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===== MULTI-ORGAN BIOPHYSICAL CONSTRAINT MATRIX =====

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===== Organ / Compartment Mass (g) Volume (L) $d\text{ (cm)}$ $t_{\max}$ @ -2°C Max Safe Rampa @ T_g =====

Bottleneck Order & Mechanism

----- Brain (Encephalon)
1400 1.350 7.00 0.74 h (44m) < 0.038 °C/min #1: Extreme ischemic/thermal limit Kidneys (Bilateral) 310 0.300 2.75 3.33
h < 0.245 °C/min #2: Endothelial osmotic vulnerability Heart (Myocardium) 330 0.320 3.00 3.12 h < 0.206 °C/min #3:
Arrhythmogenic hypothermic arrest Liver (Hepatic parenchyma) 1800 1.750 6.50 0.57 h (34m) < 0.044 °C/min #4:
Diffusion distance & microperfusion Lungs (Bilateral) 1000 0.950 5.50 1.05 h < 0.061 °C/min #5: Capillary shear &
alveolar edema Pancreas 100 0.095 1.50 10.5 h < 0.822 °C/min #6: Autolytic enzymatic activation Trunk Deep Core (Bulk)
25000 24.50 12.50 0.04 h (2.4m) < 0.012 °C/min #7: Absolute whole-body thermal barrier =====
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> ****Reading this table.**** The column d is an anatomical half-thickness estimate produced by the toolkit's organ projector. It is

7.2 The Whole-Body Conduction Barrier

The data reveals an insurmountable physical bottleneck: while small organs such as the kidneys or pancreas have critical half-thick

****Robustness of the organ-scale verdicts against the geometry premise.**** Because LC depends on an assumed shape, each verdict ca

Organ	Mass (g)	LC (cm)	Required CCR (°C/min)	Required molarity (mol/L)	Flip factor	Verdict
Pancreas	90	0.93	2.65	8.48×5.1		viable
Kidney	150	1.10	1.88	8.58×4.08		viable
Heart	300	1.38	1.19	8.69×3.24		viable
Brain	1400	2.31	0.425	8.95×1.94		viable
Liver	1500	2.37	0.406	8.96×1.90		viable
Whole body	70 000	12.78	0.014	9.80	—	not viable

The brain, the organ on which the entire clinical argument rests, has the tightest surviving margin: its verdict reverses only if

8. The Epistemic Landscape, Frontier X_2 , and the P19 Experiment

===== THE
FRONTIER X2 DILEMMA
===== Rate
used: 147 °C/min | V3: German et al. (PNAS 2026) | LTP PRESERVED (138%, $p=0.072$) | [Mouse slice scale only; $LC \leq$
0.35 mm] | FRONTIER X2 | (NO KNOWN CHEMISTRY HAS BOTH | SCALE AND FUNCTION VERIFIED) | Low
CCR (0.10 °C/min) | M22: Fahy et al. (2004, 2026) | ORGAN SCALE VITRIFICATION ($LC \geq 3.5$ cm) | [Kidney
transplant 8/8; Brain LTP UNPROVEN] +----- No Function Verified
Electrophysiology Proven
=====

8.1 The Empty Intersection

****V3 Formulation** (German et al., *PNAS* 2026): Proves that functional synaptic electrophysiology (LTP) can be preserved follow
****M22 Formulation** (Fahy et al., 2004, 2026): Proves that macro-organs ($LC > 3.5 \text{ cm}$) can be vitrified without ice. HO

The intersection between **low critical cooling rate** ($CCR \leq 0.43 \text{ }^\circ\text{C/min}$) and **verified electrophysiology**

8.2 The Decision Rule of Experiment P19 (as Frozen)

P19 asks whether M22, a chemistry that scales, preserves long-term potentiation in adult mouse
hippocampal slices (350 μm , same protocol as the V3 positive-control work). Four arms, $n = 8$ slices
from $n = 4$ animals per arm, recording blind to condition: **(A)** unvitrified control; **(B)** V3, a
replicate of the published result and the positive control; **(C)** M22 at its working concentration;
(D) loading and unloading of M22 only, without cooling, which separates chemical toxicity from thermal damage.

****Primary variable:**** LTP in SC-CA1 at 60 min after HFS, as a percentage of baseline fEPSP. The thresholds
are anchored in the published V3 result (control $157.7 \pm 7.1\%$; V3 $138.1 \pm 6.9\%$; difference not significant, $P = 0.072$)

****PASS**** if arm C reaches $\geq 130\%$ and does not differ from B (95 % CI of the difference within $\pm 25\%$ percentage points).
****FAIL**** if C $\leq 110\%$ (no useful potentiation) while B $\geq 130\%$.
Between 110 and 130 %, or with B failed: ****inconclusive****, and it is not reinterpreted.
****Quality control:**** arm B must replicate the published result within $\pm 20\%$ percentage points; otherwise the experiment is no

****Diagnosis provided by arm D.**** If D fails like C, the damage is chemical toxicity and the CPA must be redesigned
(by $\sqrt{v}$, temperature and loading protocol). If D passes and C fails, the damage is thermal or due to devitrification and is at

****Respiration is a secondary measure and not a threshold.**** Basal respiration and reserve capacity are recorded so
that results can be compared with the published values (control $173.3 \pm 6.7\%$; 65 % at 10°C , $80.4 \pm 5.6\%$).
The framework's Arrhenius derivation predicts a damage $D_{CPA} = 0.142$ in arm C, i.e. an OCR retention near
86 %; a damage above 0.25 would refute that ****derivation****, and one at or below 0.10 would confirm it. This tests the derivation
not P19: P19 is decided by LTP alone.

****What it decides.**** PASS means the physical route has an exit: a chemistry exists with demonstrated function and with a CCR

compatible with human scale (the design specification of $0.425\text{ }^{\circ}\text{C}/\text{min}$ for a human brain bottleneck is chemical, and the framework's prediction P18 would then be that partial freezing or supercooling reach the clinic before the brain is frozen).

> **Correction relative to an earlier draft.** An earlier version of this section stated a "double-threshold" rule (OCR 85% called "Metabolic OK, Plasticity Open"). That rule did **not** match the hash-locked preregistration and has been removed. The **authority**; a document that disagrees with it is the document that is wrong (rule R2).

8.3 The Five Hash-Locked Preregistrations

All falsification thresholds below were frozen **before** any data exist and are protected by SHA-256 certificates over `preregistration_date + protocol_text`, stored in `red/prereg.lock`. The engine recomputes every hash on each run and halts with an alarm if any frozen text has changed. This enforces the framework's governing rule: **a threshold may never be moved to rescue a hypothesis.** Adding a **new** preregistration is legitimate; editing a frozen one is not.

ID	Question	Primary endpoint & threshold	SHA-256 (first 16)	Status
P15	Does the connectome survive a realistic cryonics protocol?	Traceable axon/synapse fraction in blind EM segmentation		
P18	Minimum falsifiable claim of the framework	For tissue $1\text{ }^{\circ}\text{C}$, reversible pause $>24\text{ h}$ requires jointly : (a) 100% of axons survive; (b) 100% of synapses survive		
P19	Does a chemistry that scales preserve LTP?	Blind CA1 fEPSP at 60 min post-HFS; PASS if arm C 130% and with $>100\%$ of axons survive		
P20	ERL of cryopreserved brain tissue without prior fixation	Expected run length (ERL), the connectomics community's metric for cryonics		
P21	Is the cooling-law exponent below 2.34?	One-sided 95% upper bound on $n < 2.34$; FAILS if the lower bound >2.34		

> **Traceability note.** An earlier draft of this manuscript presented a set labelled "P1-P5" that did not correspond to the repository's frozen record. Those labels are withdrawn. The five preregistrations above are the complete and only preregistered set, and each is verifiable against `red/prereg.lock` by recomputing its hash from the published text.

8.4 The Design Specification: Inverting the Screening Question

Conventional CPA development benchmarks candidates against M22, the most capable published vitrification solution ($0.10\text{ }^{\circ}\text{C}/\text{min}$ at 9.345 M). This anchors the search on a **solution** rather than on a **requirement**, and it discards candidates that would in fact suffice.

The framework computes the requirement directly. Anchored on the measured litre-scale cooling datum (3 L vessel, 2.20 cm , $0.47\text{ }^{\circ}\text{C}/\text{min}$ at the centre) and the conduction exponent $n = 2$:

$$\text{CCR}_{\text{required}}(\text{LC}) = 0.47 \cdot \left(\frac{2.20}{\text{LC}}\right)^n, \quad \text{LC} = \frac{V}{A} = \frac{r^3}{\kappa_{\text{geom}}}$$

with $\kappa_{\text{geom}} = 1$ (sphere), 1.5 (cylinder), 3 (slab). For a 1400 g human brain:

$$\text{CCR}_{\text{required}} = 0.425\text{ }^{\circ}\text{C}/\text{min}$$

This is $4.25\times$ more permissive than M22. The practical consequence is a screening criterion any laboratory can apply to any candidate chemistry with a single DSC measurement:

Candidate	Measured CCR ($^{\circ}\text{C}/\text{min}$)	vs brain requirement (0.425)	Verdict
M22	0.10	clears by $\times 4.25$	ADMISSIBLE (scale)
VS55	2.5	misses by $\times 5.9$	REJECTED
VMP	5.4	misses by $\times 12.7$	REJECTED
V3	≤ 147 (the rate used; CCR not measured)	not determined	NOT DETERMINED
Low-toxicity candidates (e.g. deep-eutectic solvents)		not measured	UNMEASURED

Two cautions are essential, and the toolkit enforces both in code:

- Clearing the CCR bar says nothing about preserved function.** Admissibility is a **necessary** condition on the ice-avoidance axis only. Frontier X_{25} is precisely the observation that no chemistry has yet cleared both axes.
- An unmeasured chemistry returns `UNMEASURED`, never a pass or a fail.** We note explicitly that an earlier version of our own toolkit hardcoded a CCR of $30\text{ }^{\circ}\text{C}/\text{min}$ for deep-eutectic solvents. No such peer-reviewed value exists, and the framework's own evidence points the other way: these solvents at 50% w/v **crystallize** (crystallization onset $-26.6\text{ }^{\circ}\text{C}$), i.e. they vitrify worse than DMSO. That constant has been removed. A hardcoded value for an unmeasured quantity is the mechanism by which a framework fabricates progress.

8.5 Value of Information: Which Experiment to Run First

A measurement is worth performing when its plausible range **straddles** a decision threshold. We rank every open quantity by expected value of information (EVOI) per unit cost, where $\text{EVOI} = \min(1, 2P(\text{crossing})) (1 + \sigma)$ under a Gaussian prior:

Open quantity	Prediction	σ	Threshold	EVOI	Rel. cost	EVOI/cost	Recommendation
CCR of a low-toxicity candidate (DSC)	$0.30\text{ }^{\circ}\text{C}/\text{min}$	0.36	0.426	0.988	1.0	0.99	RUN FIRST
Cooling exponent n (P21)	2.00	0.26	2.34	0.241	0.5	0.48	RUN
Nucleation rate J at $-6\text{ }^{\circ}\text{C}$	$0.57\text{ }^{\circ}\text{C}/\text{L}\cdot\text{h}$	0.285	0.85	0.419	12	0.03	defer (cost)
Human τ_{eq} , optimal reperfusion	17 min	7.65	12.5	1.000	30	0.03	defer (cost)
P19 (LTP after vitrification)	0.142	0.078	0.25	0.179	20	0.01	defer (cost)

Three results deserve emphasis.

First, a differential scanning calorimetry run on a low-toxicity candidate is the single most valuable experiment in the framework (EVOI 0.988 at unit cost), because its plausible range straddles the $0.426\text{ }^{\circ}\text{C}/\text{min}$ bar. If it clears, frontier X_{25} closes on the scale axis **without**

executing P19**. Days of instrument time replace a full electrophysiology campaign.

Second, P21 – a control vessel, three small bags, three large bags and one thermocouple – ranks second overall, forty-eight times above P19 per unit cost. It had been deprioritized as trivial. It is not: the exponent $n = 2$ carries the entire multi-organ projection of \$7 and has never been measured.

Third, human τ_{eq} has the highest EVOI in the framework (1.000) and is deferred purely on cost. This is a statement about resources, not about relevance, and we record it as such rather than quietly omitting the most informative unmeasured quantity in the field.

9. Regulatory Pathways, Clinical Translation, and Experimental Models

=====

===== EXPERIMENTAL & REGULATORY TIER MATRIX =====

=====

===== Tier Model System Regulatory / Ethical Framework TRL Level Translational Role & Key Advantage

----- Tier 1 In Vitro / Brain Slices IACUC Standard Animal Protocol TRL 2-3 Synaptic plasticity (LTP), pure cytotoxicity Tier 2 Rodent Whole Organ IACUC Small Animal Surgery TRL 4 Vascular orthotopic transplantation, 100-day survival Tier 3 Porcine In Vivo Model Pre-clinical Large Animal Com. TRL 5 Human-scale vascular organ translation (30-50 kg) Tier 4 Surgical Discard Organ IRB Approval + UAGA Consent TRL 3-5 CRITICAL BRIDGE: Real human tissue, ex vivo normothermic Tier 5 Emergency Clinical (EPR) FDA / EMA EFIC (21 CFR 50.24) TRL 6-7 Ultra-short deep hypothermia (<45 min); resuscitation Tier 6 Post-Mortem Field Stasis UAGA Whole-Body Donation TRL 1-2 Field cryoprotection; penalized by warm ischemic delay =====

The Human Discard Organ: The Indispensable Translational Bridge

A critical methodological innovation in StasisPath is the formal inclusion of **human surgical discard organs** (Tier 4: kidneys v

10. Formal Epistemic Safeguards and No-Claim Declarations

To maintain absolute scientific integrity and prevent misinterpretation by the public or scientific community, StasisPath enforces

+-----+				
	FORMAL	NO-CLAIM	DECLARATIONS	(EPISTEMIC SAFEGUARDS)
+-----+				

| 1. Whole-Body Stasis is Open and Unvalidated: We DO NOT claim that reversible whole-body cryopreservation in adult |

| non-hibernating mammals is currently feasible. There is zero published evidence of functional recovery in an |

intact adult mammal following vitrification.
2. Multi-Organ Extrapolation is Strictly Conjectural: Success at cellular (E5) or slice scales (E2) cannot be assumed to scale to vascularized organs (E3) or whole organisms (E0). Extrapolating isolated organ kinetics to intact bodies remains an unverified hypothesis.
3. Frontier X2 Remains Open: Numerical modeling demonstrates target parameter feasibility, but X2 remains formally OPEN until direct electrophysiological verification (LTP) is achieved and replicated in pre-registered trials.

10.4 Self-Audit: Three Corrections to Our Own Results

The framework was audited against its own premises using formal regime guards, dimensional analysis, and value-of-information ranking. The audit was designed to make results *less* impressive, and it did. All three findings are reported here because a framework that only ever revises upward is not auditing itself.

(1) Q_{10} applied below the glass transition – an unphysical extrapolation. Bound C2 reported that ten-year storage requires $T < -129^\circ\text{C}$. But $T_g = -123^\circ\text{C}$: below it there is no liquid water and no metabolism, so Q_{10} is not merely extrapolated, it is **undefined**. The -129°C figure is a number without physics behind it and is withdrawn as a quantitative claim.

The *verdict* of C2 survives untouched, because what C2 actually asserts is the far weaker and robust statement that ten-year storage requires $T < 0^\circ\text{C}$ under *any* measured Q_{10} . We note separately that no Q_{10} in the framework was measured below -10°C ; everything below that temperature is projection.

(2) A silent geometry premise, now quantified. $LC = V/A$ was computed with a sphere formula ($r/3$) throughout, including for the torso, which is a cylinder ($r/2$). Rather than assert the correction is harmless, we computed the flip factor for every organ (§7.2). The brain's verdict reverses only at an LC error of $1.94\times$, against a worst-case geometric error of $1.5\times$: it survives. The whole-body verdict was already negative and the correction makes it more so (LC 8.5 $\rightarrow$ 12.8 cm, required molarity 9.60 $\rightarrow$ 9.80 mol/L). A related inconsistency between the manuscript (12.5 cm) and the computational engine (8.5 cm) has been unified in favour of the cylinder.

(3) A hidden density assumption. Converting organ mass in grams to a length in centimetres requires a density. The framework assumed $\rho = 1.0\text{ g/cm}^3$ without declaring it (tissue is ≈ 1.05 , a -1.6% bias in LC). The bias is small; the undeclared premise was not acceptable. Density is now an explicit parameter of the public API.

A fourth item is a methodological disclosure rather than a physical finding. The exponent $n = 2$ underlying the entire multi-organ projection of §7 is preregistered (P21) but *has never been measured*. Every number in the organ table inherits that status. We flag it on every computation rather than in a footnote.

11. The `stasispath` Computational Toolkit

Alongside the theoretical framework, we release *stasispath*, an open-source, zero-dependency Python 3 package (`pip install .`)

design_spec – the inverted screening question of §8.4. `required_ccr(mass_g, geometry)` returns the cooling rate a given piece

assumptions – executable regime guards for the premises that silently invalidate cryobiology calculations: the Biot boundary

experiment_value – the EVOI-per-cost ranking of §8.5, so that any group can re-rank the open set against its own cost structure

stasispath-tools/ **pyproject.toml** # Author: Alejo Malia **README.md** # Complete biophysical guide & API docs **stasispath/** **__init__.py** # Public exports & versioning **constants.py** # Empirical anchors (Q10, Ea, E, nu, alpha) **units.py** # Strict dimensional conversions ($^\circ\text{C}$, K, bar, MPa) **f3_ischemia.py** # Piecewise Q10 integration (τ_{eq}) **f4_nucleation.py** # Poisson ice nucleation probability **f5_cooling.py** # Conduction limits & critical length scales (LC^*) **thermal_stress.py** # EN 14620-5 thermoelastic stress & fracture **cpa_toxicity.py** # Arrhenius cytotoxicity accumulation (D_{CPA}) **cpa_washout.py** # Kedem-Katchalsky osmotic washout protocol scheduler **thermal_recipes.py** # Freezer ramp generator (Planer Kryo, Asymptote) **organ_projector.py** # Du Bois / ICRP 89 anthropometric projector **boa_connector.py** # Radiological CT mask ingestor (BOA, TotalSegmentator) **predictions.py** # frozen P19 decision rule + exploratory hypotheses H2-H5 **reporting.py** # Markdown synthesis & laboratory summary generator **cli.py** # Full command-line interface **tests/** # 28/28 passing unit tests (0.005s execution)

Automated Verification

The suite comprises **79 unit tests** (`python -m pytest`, no external dependencies) and covers all analytical and epistemic modules

- * **TestF3Ischemia**: Verifies bilinear Q_{10} inflection at 15.0°C and hibernator domain flag.
- * **TestF4Nucleation**: Validates Poisson survival probabilities and interfacial defect collapse.
- * **TestThermalStress**: Confirms safe cooling rates near T_g ($< 0.150^\circ\text{C/min}$) and fast-rate fracture.
- * **TestCPAToxicity**: Validates subzero kinetic toxicity rescue ($\times 3.8$ attenuation at -22°C).
- * **TestCPAWashout**: Proves that abrupt washout causes osmotic lysis and verifies the 7-step Mannitol clamp ($\Delta V/V_0 \leq 1.1$).
- * **TestOrganProjector**: Validates Du Bois BSA calculations, ICRP 89 organ volumes, and bottleneck ranking.
- * **TestBOAConnector**: Confirms ingestion of TotalSegmentator / BOA radiologic volume masks.
- * **TestP19FrozenRule**: Verifies the P19 decision rule at every frozen threshold and every boundary (130 %, 110 %, $\pm 25\text{ pp}$, $\pm 10\text{ pp}$).
- * **TestThermalRecipes**: Validates cryogenic freezer script formatting for Planer Kryo and Asymptote systems.
- * **TestDesignSpec**: Verifies that the design specification reproduces the knowledge-graph engine to within 0.002°C/min on every organ.
- * **TestAssumptions**: Verifies that each regime guard fires on the exact conditions that produced real errors in this work – LC^* .
- * **TestExperimentValue**: Verifies that EVOI is maximal at a decision boundary and vanishes far from it, that calorimetry outranks

Open Science, Data Availability & Licensing

All code, mathematical models, parameter tables, and preregistration protocols for the *StasisPath* framework are released for academic

12. References

The framework's knowledge graph catalogues **91 sources**; the subset cited directly in this manuscript is listed below. Three citations in the previous version of this manuscript were incorrect and are corrected here: reference 7 (journal: *Nature Protocols*, not *Nature Medicine*), reference 24 (pages 966 ff., not 1082–1093), and reference 38 (*Energy* **334**:137568, not 312:133501). Full provenance for all 91 sources, including status tags and the specific numerical value each one supplies, is in `red/stasispath.yaml`.

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Data and Code Availability

The complete knowledge graph (`red/stasispath.yaml`, 91 sources, 251 nodes), the verification engine (`red/motor.py`), the self-audit module (`red/sfsa_auditoria.py`), the hash-locked preregistration certificates (`red/prereg.lock`), and the `stasispath` toolkit with its 79 unit tests are released together. Every component is released under a Creative Commons Attribution-NonCommercial-ShareAlike license. The engine; none is hand-edited. Readers who wish to verify the preregistrations may recompute each SHA-256 from the published protocol text and the preregistration date.

Note on the frozen record. The preregistration P18 retains the project's prototype name ("DSNY") inside its frozen text. It has deliberately **not** been renamed, because editing a hash-locked preregistration – even cosmetically – would destroy the evidence that it was never altered. The prototype name survives there and nowhere else.

Contributors

Alejo Malia conceived the research question and the MATE + TRIADA method, directed the work, and is responsible for every claim in this paper.